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| <b>12</b> | <b>A</b> | <p>By conservation of energy, the loss of heat energy store by glass stopper is the gain in the heat energy store by the water.</p> $m_w c_w \Delta \theta_w = m_g c_g \Delta \theta_g$ $0.160 \quad 4200 \quad 28 - 19 = 0.100 \quad c_g \quad 100 - 28$ $c_g = 840 \text{ J kg}^{-1} \text{ K}^{-1}$ |
| <b>13</b> | <b>B</b> | Work is done on the air through compression by the pump, increasing its                                                                                                                                                                                                                                |